

Tekkscope

Full-Stack
AI Research Platform

Search · Browser Extraction · Multi-Provider LLMs · Streaming · MCP · Reports

Core research platform ownership

8

ordered LLM providers

sync · async · tool calls

7

MCP JSON-RPC tools

stdio agent surface

8

streaming workers

Redis cross-worker delivery

3

report formats

PDF · DOCX · Markdown

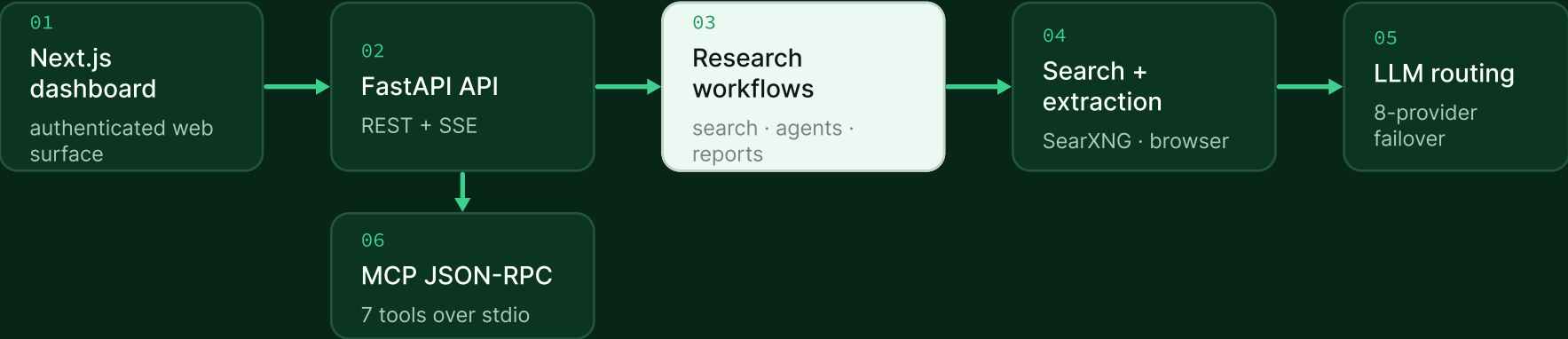
ENGINEERING NARRATIVE

One shared research core powers the web dashboard, public API, and MCP surface.

Built the FastAPI research API, ordered provider failover, SearXNG retrieval, Crawlee/Playwright extraction, SSE streaming, Redis Pub/Sub, MCP tools, reports, structured JSON, Supabase-backed product infrastructure, deployment configuration, and backend tests.

SHARED RESEARCH CORE

Three access surfaces converge on the same source-backed research system.



ENGINEERING SCOPE

- Research API
- Browser Extraction
- LLM Routing
- SSE / Redis
- MCP
- Reports / Structured JSON

CORE TECHNOLOGIES

Python · FastAPI · LangChain · LangGraph · SearXNG · Crawlee · Playwright · Redis

Supabase · Next.js · React · Docker · Kubernetes · AKS · GitHub Actions CI/CD

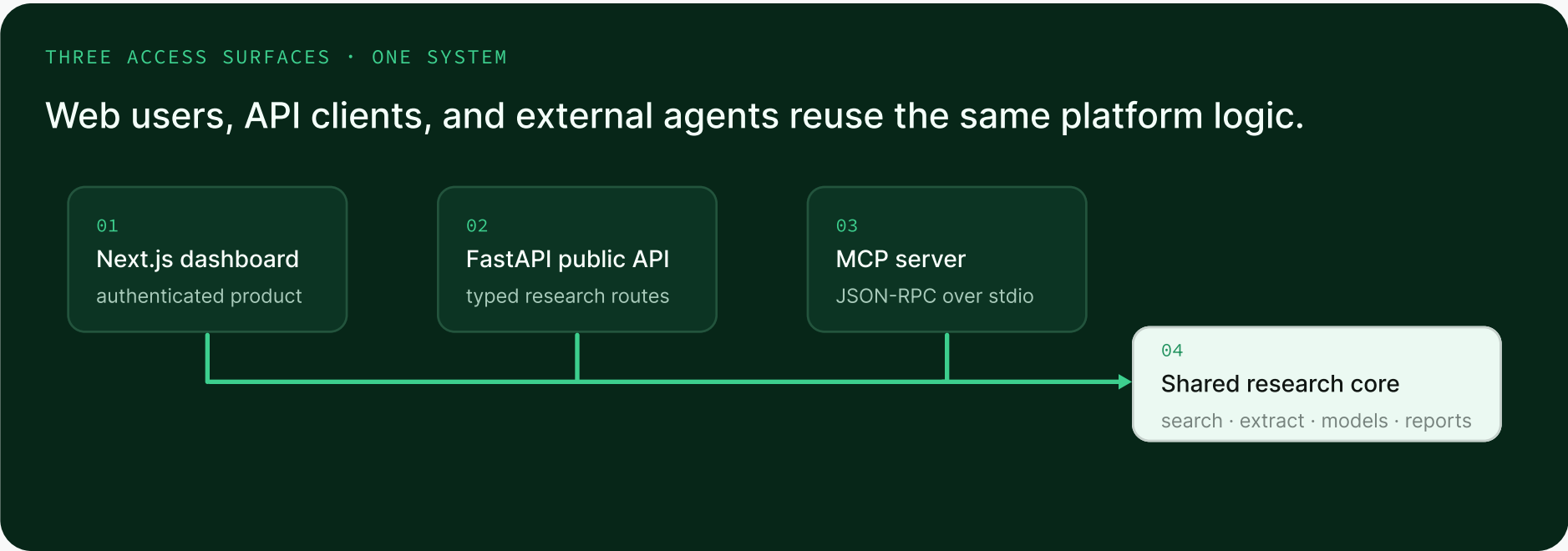
Product Surfaces & Engineering Ownership

Shared core behind browser, API, and MCP workflows.

MY ENGINEERING OWNERSHIP

Built the core Tekkscope research platform.

FastAPI research routes · model routing · retrieval/extraction · SSE/Redis · MCP · reports/NLTS · Supabase data flows · deployment and backend tests.



OWNERSHIP LEDGER

PRIMARY OWNER	Research API typed routes + workflow orchestration	PRIMARY OWNER	Model routing ordered failover · sync / async / tools
PRIMARY OWNER	Search retrieval SearXNG · query expansion · dedupe	PRIMARY OWNER	Browser extraction Crawlee / Playwright · bounded fan-out
PRIMARY OWNER	Real-time streaming SSE · asyncio queues · Redis Pub/Sub	PRIMARY OWNER	Agent surface MCP JSON-RPC server · 7 tools
PRIMARY OWNER	Artifacts & data reports · NLTS · Supabase persistence	PRIMARY OWNER	Delivery & tests Docker · AKS · CI/CD · backend tests

PRIMARY CAPABILITIES

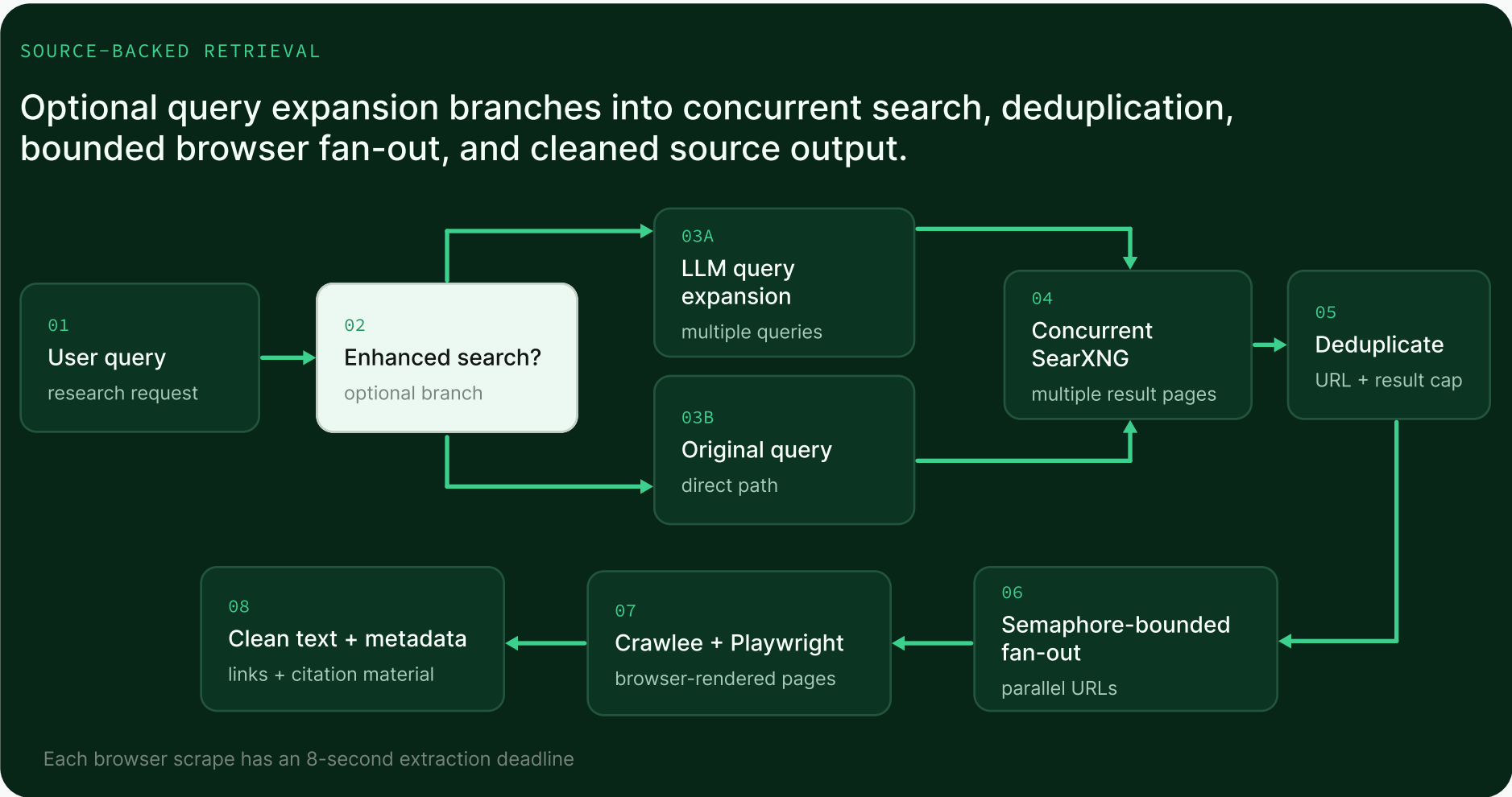
- Link Search
- Extracted Content
- Lens
- DeepLens
- ReportLens
- Tek Chat
- NLTS

Seven primary capabilities share retrieval, extraction, model, streaming, storage, and artifact primitives rather than duplicating platform logic.

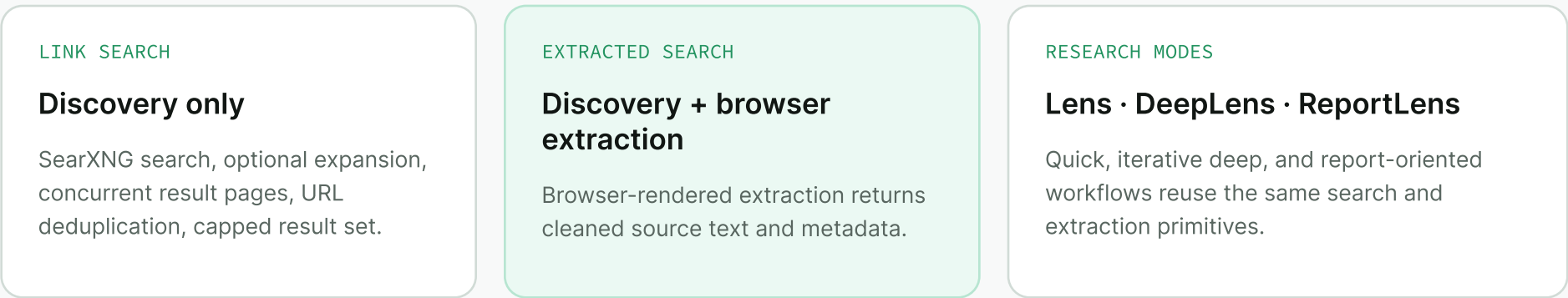
Search, Extraction & Research Pipeline

Source discovery and browser-rendered extraction are shared primitives across research modes.

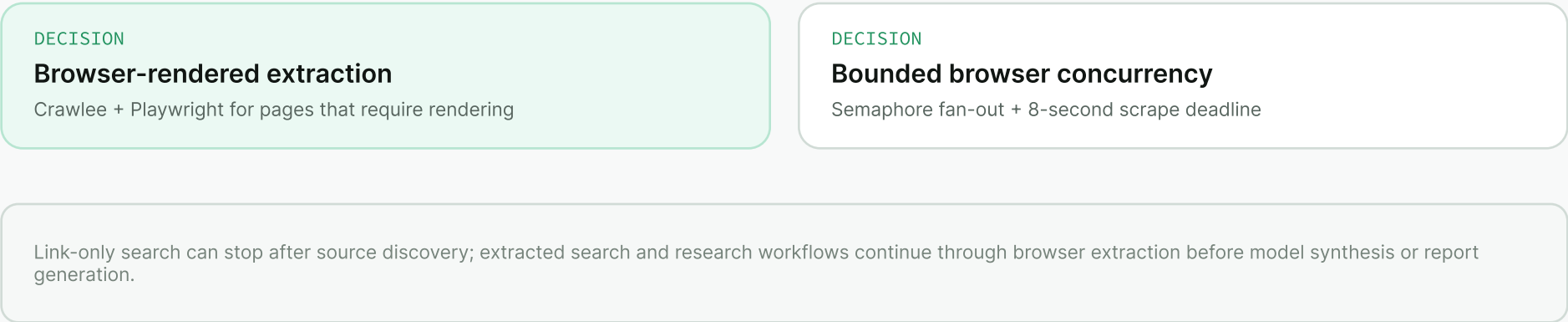
RETRIEVAL PIPELINE



REUSED ACROSS MODES



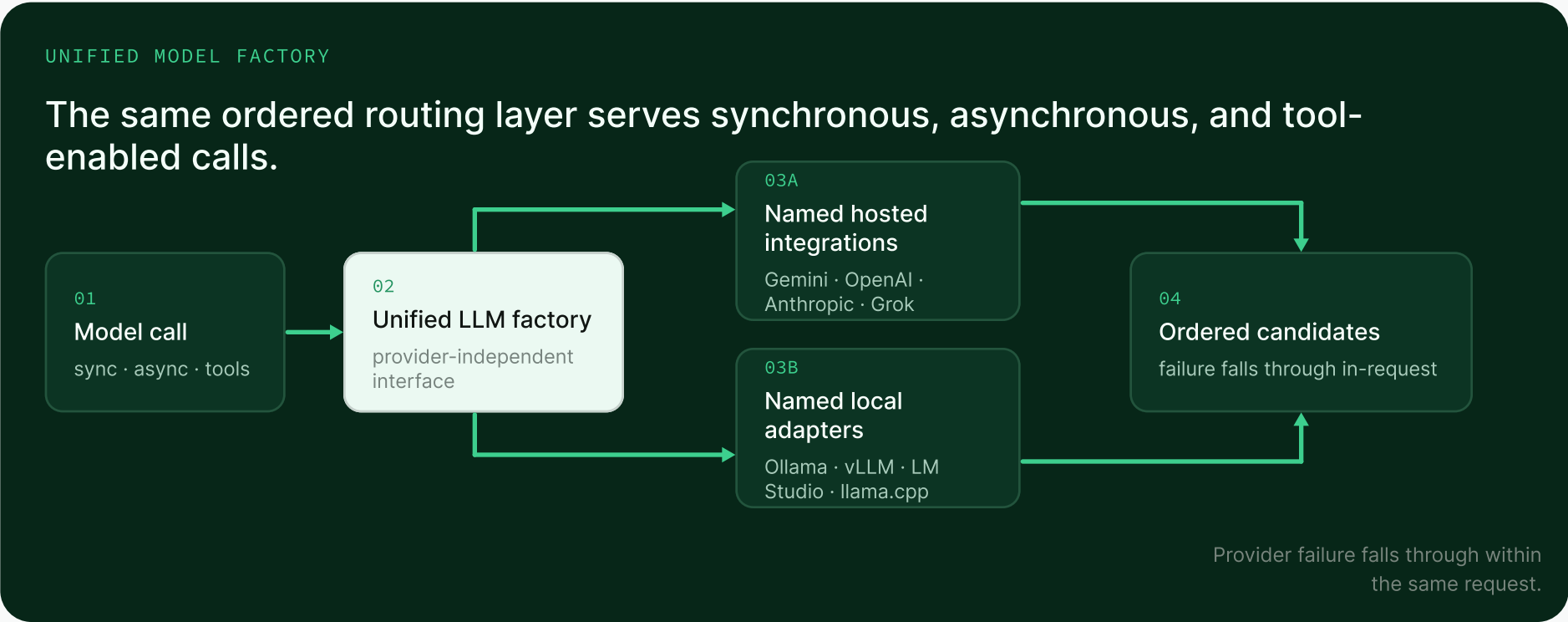
ENGINEERING DECISIONS



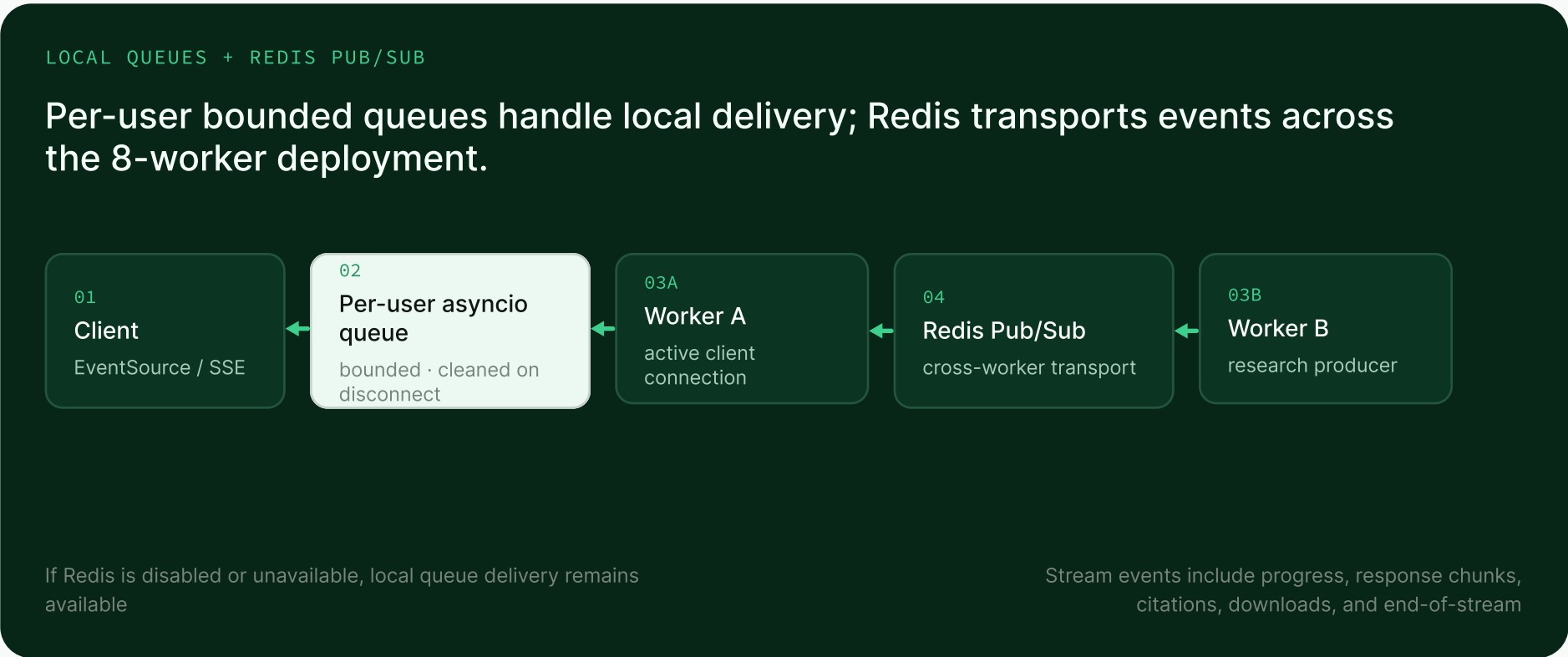
LLM Routing, Failover & Real-Time Streaming

Provider resilience and streamed execution are separate layers joined inside one research request.

ORDERED MODEL ROUTING



REAL-TIME SSE STREAMING

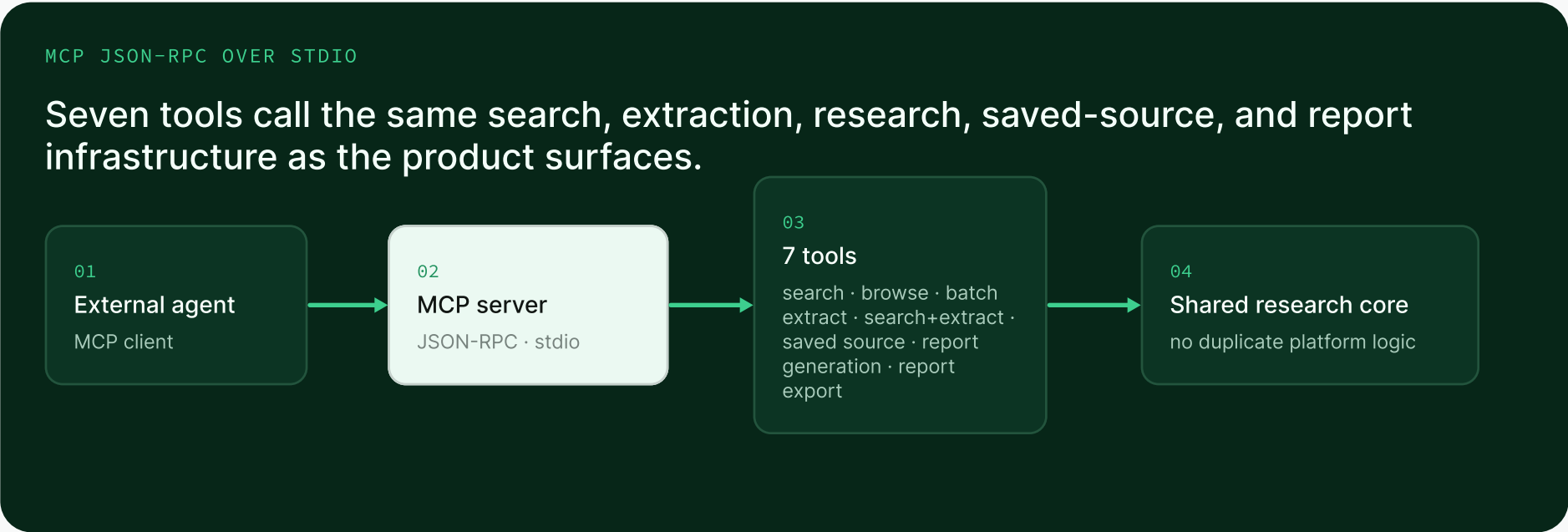


Production streaming runs across 8 workers. Redis enables cross-worker event delivery; message de-duplication and bounded queues are part of the connection manager.

Agent Access, Structured Output & Stored Research Artifacts

The same core serves external agents, machine-consumable JSON, and downloadable long-form reports.

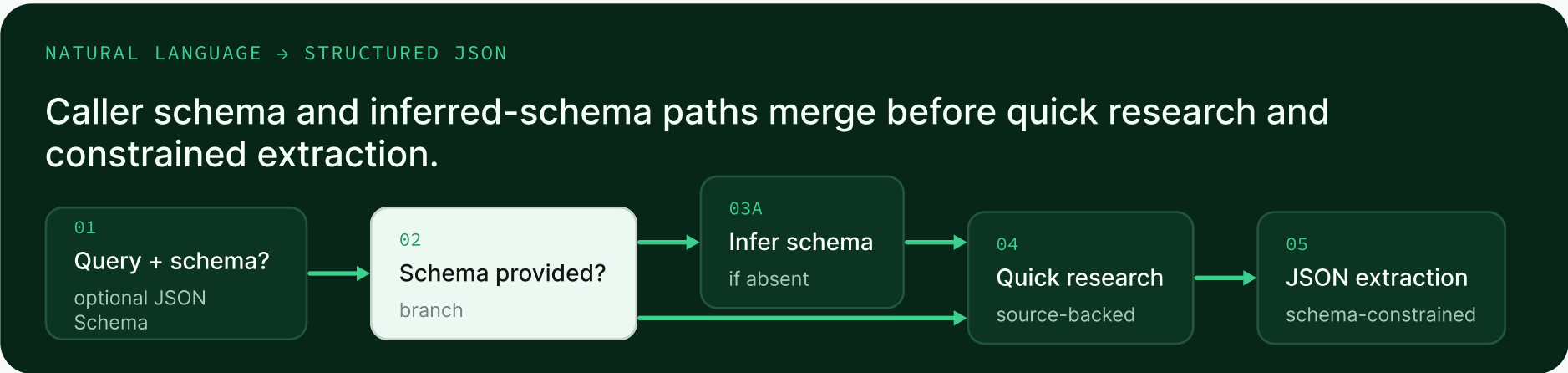
MCP AGENT SURFACE



REPORT GENERATION



STRUCTURED OUTPUT

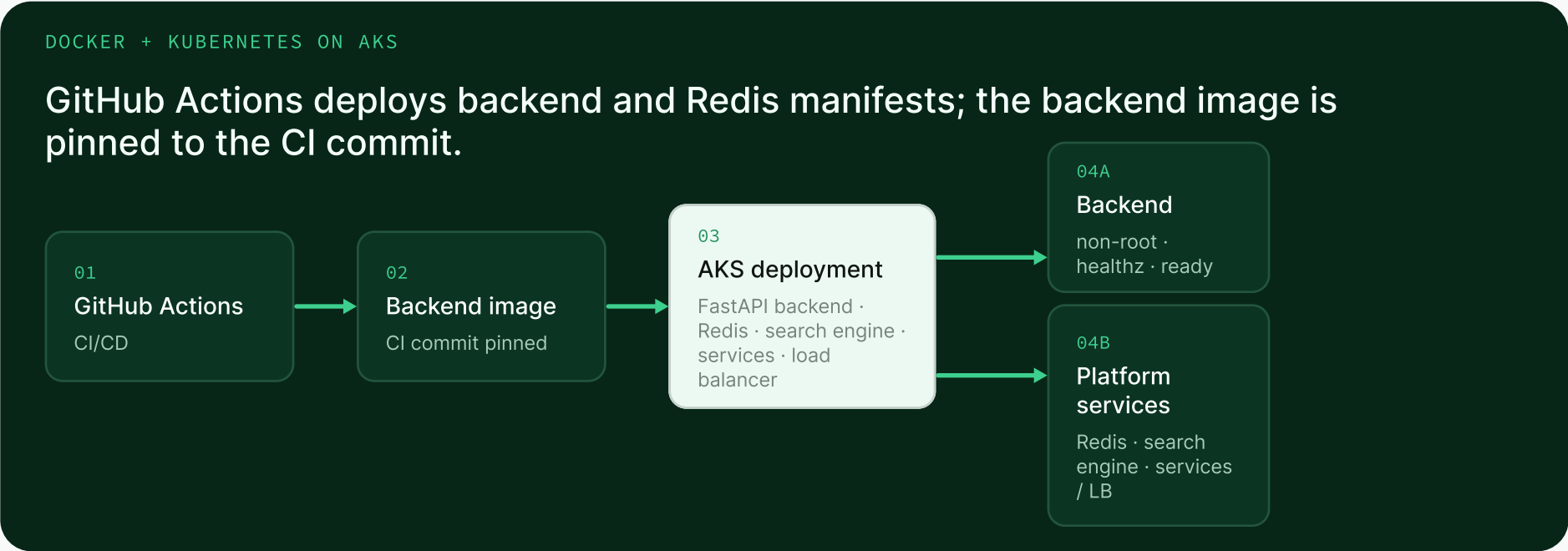


NLTS returns structured_output, schema_used, and research_summary. Report workflows persist generated artifacts and create downloadable URLs.

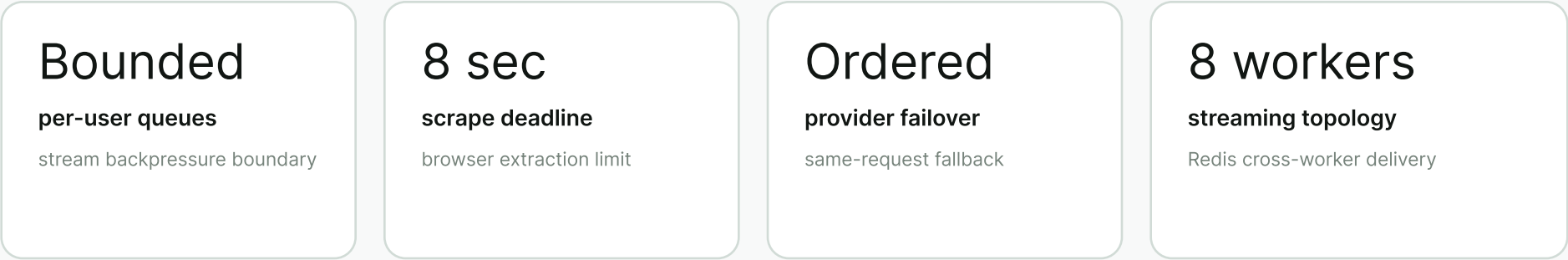
Infrastructure, Reliability & Engineering Decisions

Deployment topology, reliability mechanisms, and the explicit factual boundary of the public case study.

PRODUCTION DELIVERY



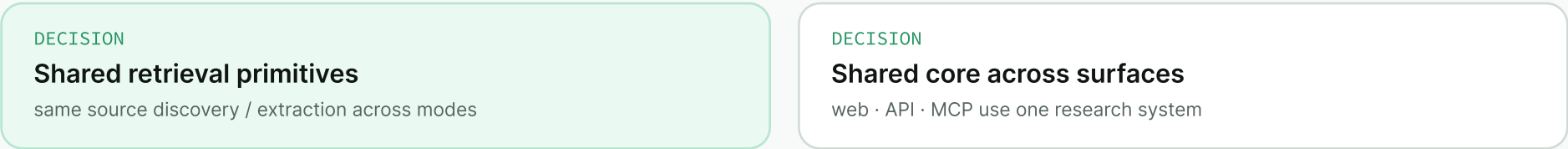
RELIABILITY MECHANISMS



VERIFIED RESULTS / FACTS

SYSTEM	VERIFIED FACT	SOURCE FRAMING
Model routing	8-provider ordered failover	sync · async · tool-enabled calls
Streaming	8-worker SSE topology	Redis Pub/Sub + local queue fallback
MCP	7 JSON-RPC tools over stdio	shared underlying research infrastructure

ENGINEERING DECISIONS



Public artifact boundary · high-level architecture and named technologies only; no private secrets, customer data, unverified performance, uptime, latency, model-quality metrics, or unnamed components/services.